

SCORED CONSISTENCY DISTILLATION OF SD3.5-MEDIUM WITH A DECODE-FREE PHOTOGRAPHIC REWARD

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1 METHOD

We distil SD3.5-Medium, a 2.5B-parameter rectified-flow transformer, into a 4-step student with no classifier-free guidance. Consistency distillation (CD) trains the student to reproduce the teacher’s sampling trajectory. The naive recipe rolls out one teacher trajectory per caption and regresses onto it. Our recipe changes two things and nothing else (Figure 1):

1. **Scored trajectory selection.** For every training caption the frozen teacher produces four candidate trajectories, each is decoded and compared with the real photograph the caption was written for, and the student is trained only on the candidate that is closest to the photograph.
2. **A decode-free photographic reward.** During training a small projector network P_ϕ maps the student’s own clean-latent prediction straight into the image-embedding space used for scoring, so a photographic similarity reward is available on every update without decoding. Because the student’s predictions drift away from the teacher’s candidates that P_ϕ was pretrained on, P_ϕ is briefly refit every 100 updates on decoded recent predictions.

Both parts use the same quantity, the DINOv2 similarity between an image and the caption’s photograph; neither reads the caption text, a VLM or a learned preference model. At inference nothing remains but a single 4-step network.

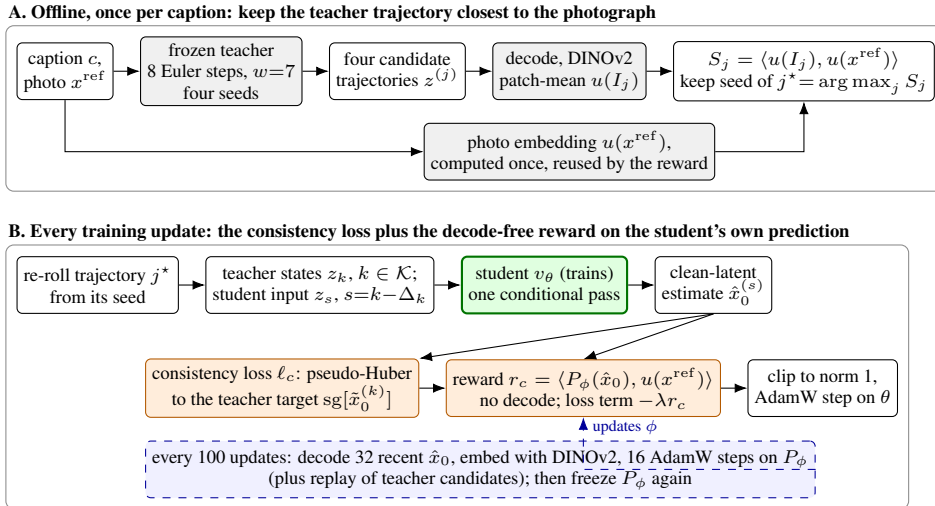


Figure 1: Overview. (A) Before training, each caption’s four teacher trajectories are decoded, embedded with a frozen DINOv2, scored against the caption’s real photograph, and only the seed of the best one is kept. (B) Each update re-rolls that trajectory, forms the student’s clean-latent estimate from a noisier state, and minimises the consistency loss minus a reward computed directly on the latent by the projector P_ϕ ; no image is decoded on the training path. The dashed block refits P_ϕ every 100 updates. Grey blocks are frozen; only the student (green) trains.

1.1 RECTIFIED-FLOW TEACHER AND CONSISTENCY DISTILLATION

The teacher is a rectified-flow model with frozen weights θ_T , operating in the D -dimensional latent space of a frozen VAE ($D=65,536$ at 512×512 pixels: a $16 \times 64 \times 64$ latent). A clean latent $x_0 \in \mathbb{R}^D$ and Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$ define a straight path $z_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$, $\sigma \in [0, 1]$, and the network predicts the velocity $v = \varepsilon - x_0$ conditioned on the caption c . Sampling is Euler integration on a σ grid, $z_{k+1} = z_k + (\sigma_{k+1} - \sigma_k) v_\theta(z_k, \sigma_k, c)$, with classifier-free guidance $v^{\text{cfg}} = v_\theta(z, \sigma, \emptyset) + w(v_\theta(z, \sigma, c) - v_\theta(z, \sigma, \emptyset))$ at $w = 7$ for the teacher. The student v_θ is a full copy of the same transformer, initialised from θ_T , and is the only network that trains; the teacher, the text encoders and the VAE decoder Dec stay frozen. There is no momentum target and no exponential moving average: every training target below is a deterministic function of the frozen teacher.

Consistency loss. Let $z_0, \dots, z_8$ be a teacher trajectory (8 Euler steps at $w=7$). Training supervises a window of states $\mathcal{K} = \{3, 4, 5, 6, 7\}$ on the 8-step grid. For $k \in \mathcal{K}$, the teacher’s realised chord across its own step and the clean-latent estimate it implies are

$$\bar{v}_k = \frac{z_{k+1} - z_k}{\sigma_{k+1} - \sigma_k}, \quad \tilde{x}_0^{(k)} = z_k - \sigma_k \bar{v}_k, \quad (1)$$

exact algebra for a straight path. The student is placed at a *noisier* state than the one the target was read from: for each k we draw $\Delta_k \sim \mathcal{U}\{1, 2, 3\}$ afresh on every visit, set $s = k - \Delta_k$, and the student forms its own clean-latent estimate with a single conditional forward pass and no guidance,

$$\hat{x}_0^{(s)} = z_s - \sigma_s v_\theta(z_s, \sigma_s, c). \quad (2)$$

The consistency loss is a pseudo-Huber distance in x_0 -space, averaged over the window:

$$\ell_c(\theta) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \rho(\hat{x}_0^{(k-\Delta_k)} - \text{sg}[\hat{x}_0^{(k)}]), \quad \rho(u) = \sqrt{\|u\|_2^2 + c_h^2} - c_h, \quad (3)$$

with $\text{sg}[\cdot]$ a stop-gradient and $c_h = 0.00054\sqrt{D} \approx 0.138$. This loss is what turns an 8-step teacher into a 4-step student; nothing below changes it. Because the student is trained with a single conditional pass against $w=7$ teacher trajectories, guidance is absorbed into its prediction and the student is sampled at $w=1$. What our method changes is *which* trajectory the student is trained on, and what, besides matching that trajectory, its own predictions are asked to do.

1.2 SCORED TRAJECTORY SELECTION

For every training caption c with reference photograph x^{ref} (its COCO source image), we draw $N = 4$ candidate teacher trajectories from four seeds fixed by the caption index and score each against the photograph. Candidate j is an 8-step Euler rollout of the teacher at guidance $w = 7$ from noise $\varepsilon^{(j)}$; only the seed needs to be stored, since a candidate can be re-rolled exactly whenever it is needed. Each candidate is decoded, $I_j = \text{Dec}(z_8^{(j)})$, and embedded with a frozen DINOv2-B vision transformer: we drop the CLS token, mean-pool the $P=256$ patch tokens and ℓ_2 -normalise,

$$u(I) = \frac{\frac{1}{P} \sum_{p=1}^P h_p(I)}{\left\| \frac{1}{P} \sum_{p=1}^P h_p(I) \right\|}, \quad S_j = \langle u(I_j), u(x^{\text{ref}}) \rangle, \quad j^* = \arg \max_{j \in \{1, \dots, N\}} S_j. \quad (4)$$

$u(x^{\text{ref}})$ is computed once per caption and reused by the reward below. Every subsequent visit of caption c trains only on trajectory j^* . Selection is computed once, offline, before training starts; no gradient flows through the scorer. The naive CD baseline replaces j^* with a uniform draw over $\{1, \dots, N\}$, stored so that every training seed sees the same draw: the two recipes see identical candidate pools and differ only in which trajectory each caption trains on and in the reward term.

1.3 A DECODE-FREE PHOTOGRAPHIC REWARD

Selection only changes which of the teacher’s trajectories the student imitates; it never moves the student’s own predictions toward the photograph directly. The reward does, and it does so without decoding a training image: a projector $P_\phi : \mathbb{R}^{16 \times 64 \times 64} \rightarrow \mathbb{R}^{768}$ maps a latent directly into the DINOv2 patch-mean space of Equation 4, so a reward can be computed and differentiated on the student’s own latent prediction with no VAE decode and no DINOv2 forward pass on the training path.

Projector. P_ϕ is a 10.7M-parameter convolutional encoder (five blocks, widths 64, 128, 256, 512, 512, the last four downsampling by 2, global-average-pooled) followed by a two-layer MLP head (512→1024→768) and ℓ_2 -normalisation. It is pretrained once, offline, on 22,000 captions disjoint from every pool it later scores (Algorithm 2 in Appendix A): for each cached candidate z_j with true embedding $e_j = u(\text{Dec}(z_j))$ and true score S_j ,

$$\mathcal{L}_{\text{pre}}(\phi) = \text{mean}_j [1 - \langle P_\phi(z_j), e_j \rangle] + \text{KL}(\text{softmax}(S/\tau) \parallel \text{softmax}(\hat{S}/\tau)), \quad (5)$$

where $\hat{S}_j = \langle P_\phi(z_j), u(x^{\text{ref}}) \rangle$ is the projector’s own score and $\tau=0.04$; the first term matches the embedding, the second preserves the within-caption ranking of the four candidates. On held-out captions the pretrained projector picks the same candidate as the decoded scorer 49% of the time (chance 25%).

Reward. On every update, the student’s clean-latent estimates at the two least-noisy supervised states are scored by P_ϕ directly:

$$r_c(\theta; \phi) = \frac{1}{2} \sum_{i=1}^2 \langle P_\phi(\hat{x}_0^{(s_i)}), u(x^{\text{ref}}) \rangle, \quad \mathcal{L}(\theta) = \mathbb{E}_c[\ell_c(\theta)] - \lambda \mathbb{E}_c[r_c(\theta; \phi)]. \quad (6)$$

The gradient of r_c flows through the frozen P_ϕ into θ only. We use $\lambda=80$, chosen so that the raw reward gradient is about 20% of the raw consistency gradient at initialisation (a 40-update probe measures a norm ratio of 0.22); the sum is then clipped to global norm 1, so λ sets the reward’s share of the update direction rather than an absolute step size.

Periodic refresh. P_ϕ was pretrained on the teacher’s candidates, not on the evolving predictions of a 4-step student, so its accuracy on what it actually scores during training is not guaranteed. Every 100 updates P_ϕ is therefore unfrozen and refit to the true DINOv2 embeddings of the student’s 32 most recent predictions (decoded only for this refresh, not for the reward), together with a replay term on a fresh batch of the original teacher-candidate data so it does not forget how to rank them:

$$\mathcal{L}_{\text{refresh}}(\phi) = \text{mean}[1 - \langle P_\phi(\hat{x}_0), u(\text{Dec}(\hat{x}_0)) \rangle] + \mathcal{L}_{\text{pre}}(\phi) \text{ on a replay batch}, \quad (7)$$

16 AdamW steps at learning rate 10^{-4} each time, after which P_ϕ is frozen again. Section 3.2 compares this schedule with a frozen projector and with lighter refreshes.

1.4 TRAINING PROCEDURE AND INFERENCE

Algorithm 1 is one optimiser step. The two decodes in the whole pipeline happen outside the gradient path: once per caption before training (selection, panel A of Figure 1) and on 32 latents every 100 updates (the refresh). At inference the student is sampled with 4 Euler steps at $w=1$: no guidance, no candidate generation, no projector. The trained student is a single network; nothing about candidate pools, scorers, the projector or the reward survives into the deployed model.

2 EXPERIMENTAL SETUP

Data. Training captions come from COCO train2017, each paired with the photograph it was written for; the photograph is used only by the scorer and the reward and never enters the student’s input. All experiments below use a pool of 3,000 captions drawn once with a fixed seed. Evaluation uses a separate, sealed pool built once with a fixed seed; fidelity references come from COCO val2017, so no evaluation caption or photograph is seen in training.

Training configuration. Both recipes use identical code, data order and hyperparameters (Table 1); the only differences are the selected trajectory and the reward term. Every reported student is the uniform weight average of its 2,000-, 4,000- and 6,000-update checkpoints, which scores 0.015–0.02 CompBench above the raw final checkpoint for both recipes and removes most of the checkpoint-to-checkpoint noise of a short schedule. Three training seeds (0–2) are run per recipe; the two recipes share seeds, data order and Δ draws at equal seed, so every comparison is seed-paired.

Algorithm 1 One training update of the student (repeat until the schedule ends)

Require: student v_θ (initialised from θ_T); frozen teacher v_T , decoder Dec and DINOv2 $u(\cdot)$; pre-trained projector P_ϕ

per-caption seed j^* and embedding $u(x^{\text{ref}})$ from selection; replay buffer of teacher candidates; $\mathcal{K} = \{3, \dots, 7\}$; $\lambda=80$

- 1: pick a caption c ; regenerate its selected trajectory $z_0, \dots, z_8$ from the stored seed $\triangleright$ no decode
- 2: **for** $k \in \mathcal{K}$ **do** $\triangleright$ five supervised noise levels, batched into one student forward pass
- 3: draw $\Delta_k \sim \mathcal{U}\{1, 2, 3\}$; $s \leftarrow k - \Delta_k$
- 4: $\hat{x}_0^{(s)} \leftarrow z_s - \sigma_s v_\theta(z_s, \sigma_s, c)$; $\tilde{x}_0^{(k)} \leftarrow \text{sg} [z_k - \sigma_k (z_{k+1} - z_k) / (\sigma_{k+1} - \sigma_k)]$
- 5: **end for**
- 6: $\ell_c \leftarrow \frac{1}{|\mathcal{K}|} \sum_k \rho(\hat{x}_0^{(k-\Delta_k)} - \tilde{x}_0^{(k)})$ $\triangleright$ Equation 3
- 7: $r_c \leftarrow \frac{1}{2} \sum_{i=1}^2 \langle P_\phi(\hat{x}_0^{(s_i)}), u(x^{\text{ref}}) \rangle$ for the two least-noisy s $\triangleright$ Equation 6; no decode
- 8: $g \leftarrow \nabla_\theta [\ell_c - \lambda r_c]$; clip g to global norm 1; AdamW step on θ $\triangleright \phi$ is not updated here
- 9: **if** the update index is a multiple of 100 **then** $\triangleright$ refresh, Equation 7
- 10: decode the 32 most recent $\hat{x}_0$ and embed them with $u(\text{Dec}(\cdot))$; sample a replay batch of teacher candidates
- 11: 16 AdamW steps on ϕ ($\text{lr } 10^{-4}$) minimising $\mathcal{L}_{\text{refresh}}$; freeze ϕ again
- 12: **end if**

Table 1: Training configuration (3k pool). The naive CD baseline and our recipe share every entry.

Captions / updates	3,000 / 6,000 (2 passes), one caption per update, 1 GPU
Optimiser	AdamW, $\text{lr } 10^{-5}$, 300 warm-up steps, weight decay 0, global grad-norm clip 1
Precision	fp32 master weights, bf16 autocast, gradient checkpointing
Teacher / candidates	SD3.5-Medium, 8 Euler steps, $w=7$; $N=4$ candidates per caption
Consistency loss	pseudo-Huber in x_0 -space, window $\mathcal{K} = \{3, \dots, 7\}$, $\Delta_k \sim \mathcal{U}\{1, 2, 3\}$
Reward (ours only)	$\lambda=80$ on the two least-noisy states; refresh every 100 updates, 16 steps, $\text{lr } 10^{-4}$
Reported weights	uniform average of the 2k / 4k / 6k checkpoints; 3 seeds per recipe

Evaluation. Alignment is measured with T2I-CompBench (8 categories: color, shape, texture, 2D-spatial, 3D-spatial, numeracy, non-spatial, complex; 2,398 sealed prompts, one image per prompt, the official per-category scorers, reported as the unweighted mean of the eight categories) and GenEval2 (800 sealed prompts of 3–10 atomic conditions over object, count, attribute, position and verb skills; a prompt scores the geometric mean of its atom scores, the official number is the mean over prompts, $\times 100$). Fidelity is measured with CMMD, precision, recall and FID against 5,000 held-out COCO val2017 photographs. Every arm and every seed is evaluated on exactly the same prompts and initial noise, so per-prompt comparisons are like-for-like. All students are sampled with 4 Euler steps at $w=1$.

Statistics. Every contrast is a seed-paired difference (seed s of ours minus seed s of naive CD), reported as mean $\pm$ standard deviation over the three seeds with the paired t -test p -value. Three seeds is a small sample: p -values below 0.05 are reported as resolved, everything else as unresolved, and every table shows the per-seed spread so the reader can judge.

3 RESULTS

3.1 MAIN RESULT: SCORED DISTILLATION WITH THE PROJECTOR REWARD BEATS NAIVE CD

Table 2 and Figure 2 give the headline comparison. On T2I-CompBench our student scores 0.4887 ± 0.0025 against 0.4738 ± 0.0017 for naive CD: a seed-paired gain of $+0.0149 \pm 0.0017$ ($p=0.004$), positive on every seed (per-seed $+0.0146$, $+0.0134$, $+0.0167$). On GenEval2 it scores 23.5 against 22.5 ($+1.00 \pm 0.91$), positive on two seeds and flat on the third, not resolved at three seeds. For scale, the same 4-step budget without distillation (the base model at 4 steps, $w=7$) reaches only 0.256 CompBench, and the 28-step guided teacher reaches 0.505: the student closes most of the gap between a 4-step and a 28-step sampler, and our recipe recovers about half of the remaining distance to the teacher over naive CD.

Table 2: Alignment on the sealed evaluation pools. Both students: 3k captions, three training seeds, averaged checkpoints, one image per prompt, 4 steps, $w=1$. Δ is the seed-paired difference to naive CD (mean $\pm$ sd over seeds, paired t -test p). Reference rows are single deterministic runs of the undistilled model.

Model	CompBench	Δ vs. naive (p)	GenEval2 ($\times 100$)	Δ vs. naive (p)
Naive CD (random trajectory, no reward)	0.4738 $\pm$ 0.0017	–	22.53 $\pm$ 0.95	–
Ours (scored trajectory + projector reward)	0.4887 $\pm$ 0.0025	+0.0149 $\pm$ 0.0017 (0.004)	23.54 $\pm$ 1.56	+1.00 $\pm$ 0.91 (0.20)
Base model, 4 steps, $w=7$ (no distillation)	0.2557	–	7.32	–
Teacher, 8 steps, $w=7$ (the distillation source)	0.4544	–	17.58	–
Teacher, 28 steps, $w=7$	0.5053	–	17.05	–

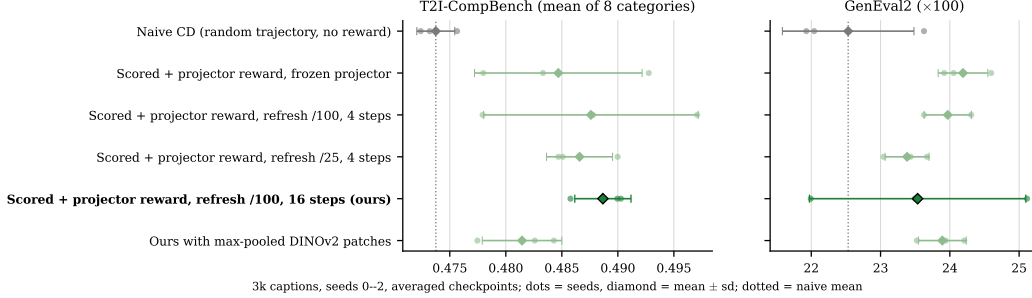


Figure 2: Every recipe of this paper on the two alignment benchmarks (3k captions, seeds 0–2, averaged checkpoints). Dots are training seeds, diamonds the mean $\pm$ sd, the dotted line the naive CD mean. Our recipe (bold) is the only one whose three seeds all sit clear of the naive CD spread on CompBench.

Where the gain is. Figure 3 and Table 3 split the CompBench gain by category. The improvement is broad rather than concentrated: seven of the eight categories move in our favour on the seed mean, and three are resolved at three seeds, colour (+0.021, $p=0.017$), shape (+0.019, $p<0.001$) and the long-caption “complex” category (+0.012, $p=0.010$). The largest mean gain is on 2D-spatial relations (+0.032) but with a wide seed spread ($p=0.09$); texture (+0.017) and numeracy (+0.009) are also positive but unresolved. Non-spatial prompts, scored by CLIPScore, do not move for either recipe.

GenEval2 by skill and prompt complexity. Appendix B breaks GenEval2 down. The share of prompts on which at least one atom is entirely wrong (“collapsed”, geometric mean below 0.05) falls from 36.6% to 33.7% (-3.0 ± 1.4 , $p=0.07$), the position skill gains the most ($+3.7 \pm 1.7$ per atom, $p=0.06$) and attribute binding gains +1.8; object presence and counting are unchanged, and the verb skill, the weakest for every 4-step model, moves within noise (-1.5 ± 3.4).

3.2 PROJECTOR-REWARD ABLATIONS

Table 4 varies only the reward, keeping scored selection and everything else fixed, and compares every variant with naive CD. Three findings. (i) Every variant beats naive CD on the seed mean, but only the refreshed projector with 16 steps per refresh (ours) does so with a tight, resolved margin (+0.0149 $\pm$ 0.0017); the frozen projector (+0.0109 $\pm$ 0.0075) and the default 4-step refresh (+0.0138 $\pm$ 0.0093) reach a similar mean at three times the seed spread, and refreshing four times as often with 4 steps (+0.0128 $\pm$ 0.0037, $p=0.03$) does not match 16 steps at the original cadence. The refresh therefore mainly buys *consistency across seeds*, which is what makes the gain resolvable. (ii) The refresh does what it is meant to: during training the projector score of the student’s own predictions rises by 0.04–0.07 in every seed while the true DINOv2 score of the same decoded latents stays flat (-0.03 to 0.00), and the within-batch correlation between the two is 0.42 over the second half of training with the refresh against 0.36 with a frozen projector; the projector tracks the ranking of the student’s predictions but not their absolute photographic similarity, which is why we read it as a regulariser that keeps the student near the photograph-like candidates it was selected

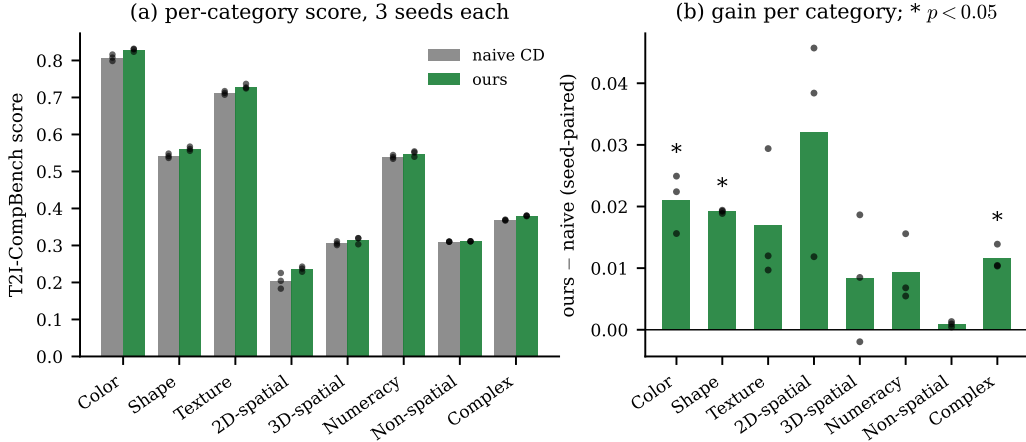


Figure 3: CompBench by category. (a) Per-category scores of naive CD and ours, dots are seeds. (b) Seed-paired gain of ours over naive CD; dots are the three per-seed differences, stars mark $p < 0.05$.

Table 3: CompBench by category, 3k pool, seeds 0–2, averaged checkpoints, one image per prompt.

Category	Naive CD	Ours	Ours – naive (p)
Color	0.8075 ± 0.0088	0.8284 ± 0.0044	$+0.0210 \pm 0.0048$ (0.02)
Shape	0.5421 ± 0.0060	0.5613 ± 0.0058	$+0.0192 \pm 0.0003$ (< 0.001)
Texture	0.7123 ± 0.0049	0.7294 ± 0.0066	$+0.0170 \pm 0.0108$ (0.11)
2D-spatial	0.2044 ± 0.0212	0.2364 ± 0.0070	$+0.0320 \pm 0.0178$ (0.09)
3D-spatial	0.3058 ± 0.0053	0.3142 ± 0.0096	$+0.0084 \pm 0.0103$ (0.29)
Numeracy	0.5391 ± 0.0052	0.5484 ± 0.0078	$+0.0093 \pm 0.0055$ (0.10)
Non-spatial	0.3103 ± 0.0006	0.3112 ± 0.0002	$+0.0009 \pm 0.0004$ (0.07)
Complex	0.3685 ± 0.0014	0.3801 ± 0.0011	$+0.0116 \pm 0.0020$ (0.01)
Mean	0.4738 ± 0.0017	0.4887 ± 0.0025	$+0.0149 \pm 0.0017$ (0.004)

on, rather than as a faithful reward. (iii) The DINOv2 pooling matters: replacing mean pooling of the patch tokens by max pooling, in both the selection score and the reward, halves the gain ($+0.0077 \pm 0.0023$). Max pooling compresses the within-caption spread of candidate scores tenfold (standard deviation 0.007 against 0.069), so at the same λ the reward gradient is ten times smaller (probe ratio 0.022 against 0.22); the selection it induces agrees with mean pooling on only 60% of captions.

3.3 FIDELITY

Alignment gains that are bought with image quality are not gains, so Tables 5 and 6 report fidelity against real photographs at both scales. At 3k the picture is mixed and small: our student is slightly closer to the real distribution on CMMD (0.817 against 0.837 ; -0.020 ± 0.026 , unresolved), its recall is higher on every seed ($+0.013 \pm 0.003$, $p=0.02$), precision is unchanged, and FID is marginally worse ($+0.47 \pm 0.14$ on a base of 30; $p=0.03$). At 118k the picture is not mixed: our student is worse than naive CD on every measure and on every seed – CMMD $+0.037 \pm 0.006$ ($p=0.008$), precision -0.014 ± 0.005 ($p=0.04$), recall -0.011 ± 0.003 ($p=0.02$), FID $+0.92 \pm 0.13$ ($p=0.007$). The alignment gain at scale is therefore paid for with a measurable loss of photographic fidelity. This is consistent with the training-time diagnostic of Section 3.2: the reward moves the student toward what the projector scores as photograph-like, and over a long schedule that is not the same as the real photograph distribution. We report it as a limitation of the projector reward rather than explain it away; a reward that tracks the true score more faithfully (Section 3.2) is the obvious remedy.

Table 4: Reward ablations, 3k pool, seeds 0–2, averaged checkpoints. Every row except the first uses scored trajectory selection with the projector reward ($\lambda=80$) and differs only in how the projector is refreshed (or, in the last row, in the DINOv2 pooling used for both selection and reward). Δ is seed-paired against naive CD. CMMD (lower is better) is the seed mean on 5,000 COCO val2017 captions.

Reward variant	CompBench	Δ vs. naive (p)	GenEval2	Δ vs. naive (p)	CMMD $\downarrow$
Naive CD (no selection, no reward)	0.4738 $\pm$ 0.0017	–	22.5 $\pm$ 0.9	–	0.837
Projector frozen	0.4847 $\pm$ 0.0075	+0.0109 $\pm$ 0.0075 (0.13)	24.2 $\pm$ 0.4	+1.66 $\pm$ 0.60 (0.04)	0.810
Refresh every 100 updates, 4 steps	0.4876 $\pm$ 0.0096	+0.0138 $\pm$ 0.0093 (0.12)	24.0 $\pm$ 0.3	+1.44 $\pm$ 1.02 (0.14)	0.843
Refresh every 25 updates, 4 steps	0.4866 $\pm$ 0.0029	+0.0128 $\pm$ 0.0037 (0.03)	23.4 $\pm$ 0.3	+0.85 $\pm$ 0.71 (0.17)	0.833
Refresh every 100 updates, 16 steps (ours)	0.4887 $\pm$ 0.0025	+0.0149 $\pm$ 0.0017 (0.004)	23.5 $\pm$ 1.6	+1.00 $\pm$ 0.91 (0.20)	0.817
Ours, max-pooled DINOv2 patches	0.4814 $\pm$ 0.0036	+0.0077 $\pm$ 0.0023 (0.03)	23.9 $\pm$ 0.3	+1.36 $\pm$ 0.99 (0.14)	–

Table 5: Fidelity on 5,000 COCO val2017 captions, 3k pool, seeds 0–2, averaged checkpoints, 4 steps, $w=1$. Contrasts are seed-paired (mean $\pm$ sd, paired t -test p).

	CMMD $\downarrow$	Precision $\uparrow$	Recall $\uparrow$	FID $\downarrow$
Naive CD	0.837 $\pm$ 0.042	0.479 $\pm$ 0.025	0.054 $\pm$ 0.003	30.42 $\pm$ 0.17
Ours	0.817 $\pm$ 0.025	0.474 $\pm$ 0.013	0.067 $\pm$ 0.001	30.89 $\pm$ 0.22
Ours – naive (seed-paired, p)	–0.020 $\pm$ 0.026 (0.32)	–0.005 $\pm$ 0.014 (0.60)	+0.013 $\pm$ 0.003 (0.02)	+0.47 $\pm$ 0.14 (0.03)

3.4 QUALITATIVE RESULTS

Figures 4 and 5 compare the two students prompt by prompt. Every column of a row starts from the same initial noise, so differences down a row are due to the model, not the draw. We also show the undistilled base model at the same 4-step budget, which is why distillation is needed at all, and the 28-step guided teacher as the ceiling. The rows are curated: for each CompBench category we ranked prompts by the ours-minus-naive CD margin averaged over the three seeds and, among the top eight, kept one whose difference is visible in the image rather than only in the evaluator’s score; Appendix C shows an uncurated random draw and the prompts on which naive CD scores higher. The pattern in the curated rows is consistent: the second object of a two-object prompt that naive CD drops appears (the black whiskers, the laptop next to the bag, the cameras with the horses, the silver car beside the truck), a fluffy or metallic material that was named becomes visible, and object counts and spatial relations that naive CD gets approximately right become exact (three turtles, five cats on the suitcase, three monkeys on the clock, the bagel to the right of the kangaroos). In the uncurated rows the two students are visually near-identical: the gain is concentrated on a minority of prompts, which is also what the per-prompt statistics say (on 21–30% of colour, shape and texture prompts our student scores at least 0.05 higher, on 11–16% naive CD does).

Figure 6 shows two 2D-spatial prompts on which our student places the objects as asked at 4, 8 and 28 sampling steps alike while the guided 28-step teacher does not: a distilled few-step student is not, skill by skill, a strictly degraded copy of its many-step teacher.

3.5 SCALING TO 118,000 CAPTIONS

To test whether the recipe survives a forty-fold larger caption pool, both recipes were trained on the complete COCO train2017 caption set (113,948 captions, two passes, 56,974 updates on 4 GPUs with one caption per GPU per update, learning rate 2×10^{-5} , 1,000 warm-up steps, three seeds each). Under this long schedule single checkpoints oscillate by up to 0.03 CompBench from one save to the next (Table 7, last row), so, as for the 3k runs, the reported student of each run is the uniform weight average of its last five checkpoints (40k, 45k, 50k, 55k and final).

Table 7 gives the result. Our recipe scores 0.4810 ± 0.0050 CompBench against 0.4668 ± 0.0036 for naive CD: $+0.0142 \pm 0.0085$ seed-paired ($+0.019$, $+0.004$, $+0.019$ on the three seeds), the same size as the 3k gain. With three seeds the seed-level test is not resolved ($p=0.10$), but the prompt-level test on seed-averaged per-prompt scores over the 2,398 prompts is ($p=4 \times 10^{-9}$). GenEval2 moves from 20.3 to 21.7 ($+1.37 \pm 1.28$; prompt-level $p=0.016$), and the share of collapsed prompts falls from 44% to 41%. Table 8 shows the gain is again broad: colour ($+0.021$, $p=0.04$), numeracy ($+0.019$), 3D-spatial ($+0.019$), shape ($+0.018$), texture ($+0.018$) and 2D-spatial ($+0.014$) all move in our favour on the seed mean, with only the CLIP-scored non-spatial category flat. What scale

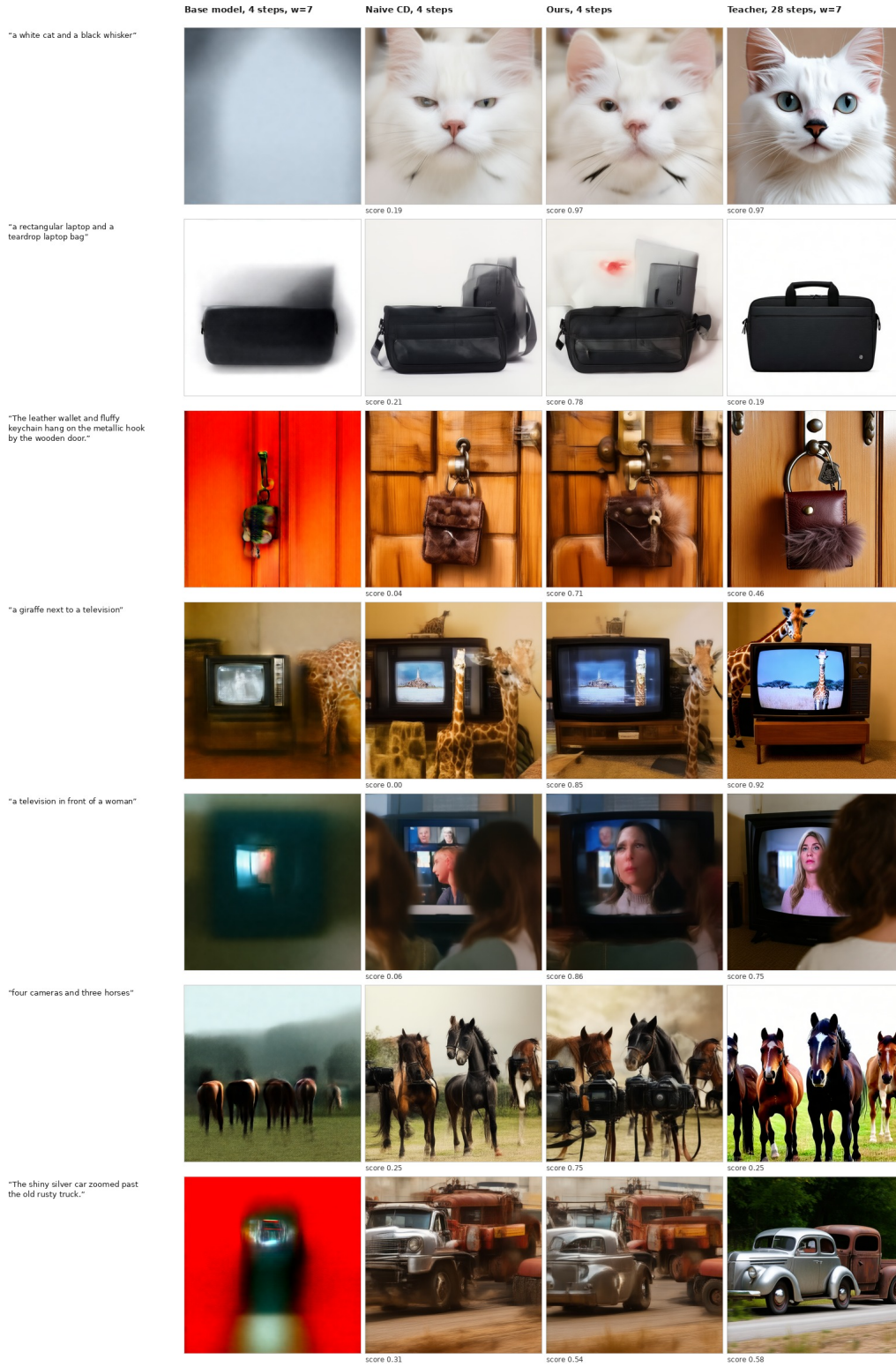


Figure 4: T2I-CompBench, one curated prompt per category (colour, shape, texture, 2D-spatial, 3D-spatial, numeracy, complex). Columns share the initial noise. Scores under the images are the official per-category evaluator’s score for that image. Base model: SD3.5-Medium sampled at the student’s 4-step budget with guidance $w=7$, no distillation.

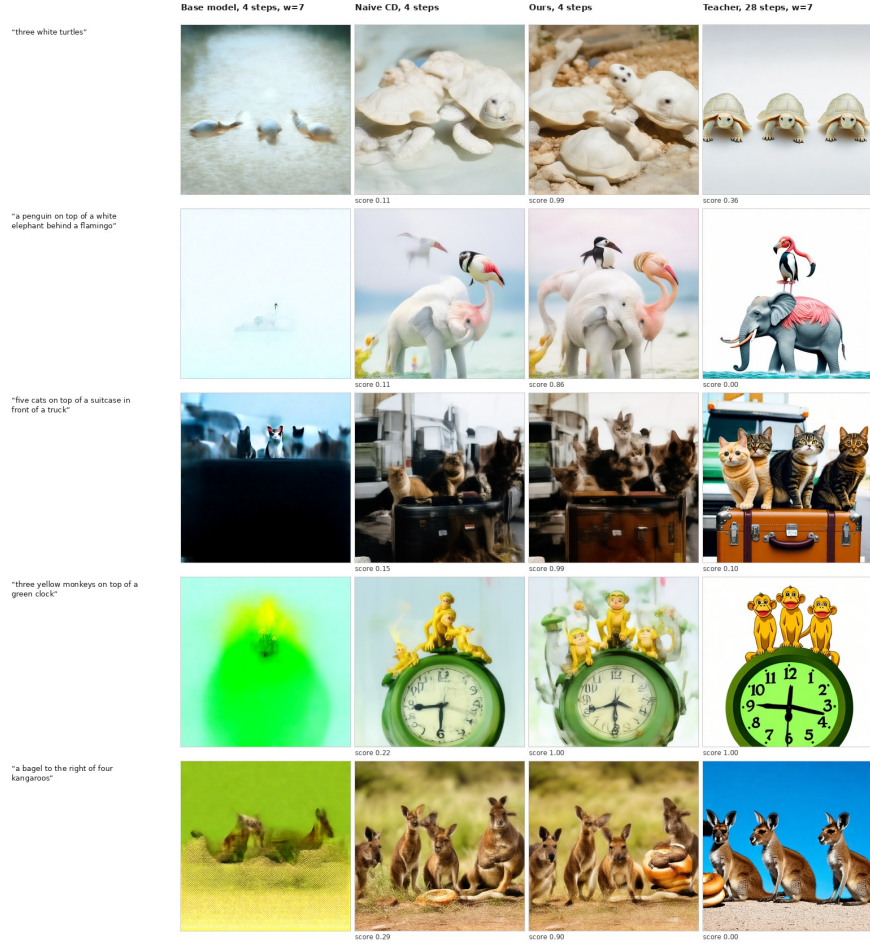


Figure 5: GenEval2, curated prompts on counting, relations and attribute binding. Scores under the images are the prompt’s geometric mean over its atoms. Columns share the initial noise.



Figure 6: Two T2I-CompBench 2D-spatial prompts, identical initial noise per prompt. Our student (no guidance) places both named objects in the requested relation at 4, 8 and 28 sampling steps alike; the frozen teacher with guidance ($w=7$) does not at 28 steps. Green borders: the requested relation is satisfied; red: it is not. Automatic 2D-spatial scores (computed for the 4-step student and the teacher only): 0.92 vs. 0.00 (top), 0.78 vs. 0.00 (bottom).

Table 6: Fidelity at 118k on the same 5,000 COCO val2017 captions, per seed (0 / 1 / 2), last-five-checkpoint averages, 4 steps, $w=1$. Contrasts are seed-paired (mean $\pm$ sd, paired t -test p).

118k, seeds 0 / 1 / 2, last-5 average	CMMD $\downarrow$	Precision $\uparrow$	Recall $\uparrow$	FID $\downarrow$
Naive CD	0.84/0.84/0.83	0.485/0.489/0.503	0.059/0.066/0.062	31.08/30.97/31.44
Ours	0.88/0.88/0.86	0.477/0.472/0.487	0.047/0.053/0.054	31.85/31.95/32.46
Ours – naive (seed-paired, p)	+0.037 $\pm$ 0.006 (0.008)	–0.014 $\pm$ 0.005 (0.04)	–0.011 $\pm$ 0.003 (0.02)	+0.92 $\pm$ 0.13 (0.007)

Table 7: 118k captions, 56,974 updates, three training seeds, last-five-checkpoint averages, 4 steps, $w=1$. Seed-paired contrasts as in Table 2; the prompt-level p is a paired t -test over the 2,398 CompBench (800 GenEval2) prompts after averaging each prompt’s score over the three seeds.

Recipe (118k captions, 3 seeds, last-5 average)	CompBench			GenEval2 ($\times 100$)		
	mean $\pm$ sd	Δ vs. naive, seed-paired (p)	prompt-level p	mean $\pm$ sd	Δ vs. naive (p)	prompt-level p
Naive CD (random trajectory, no reward)	0.4668 $\pm$ 0.0036	–	–	20.29 $\pm$ 0.48	–	–
Ours (scored trajectory + projector reward)	0.4810 $\pm$ 0.0050	+0.0142 $\pm$ 0.0085 (0.10)	< 0.001	21.66 $\pm$ 1.13	+1.37 $\pm$ 1.28 (0.21)	0.02
Raw final checkpoints (before averaging), CompBench per seed: naive 0.4573 / 0.4570 / 0.4591; ours 0.4584 / 0.4514 / 0.4718						

does *not* buy is absolute quality: the 118k students are no better than their 3k counterparts (naive CD 0.467 vs. 0.474; ours 0.481 vs. 0.489) despite ten times the updates, and their consistency loss ends at the same level. We return to this in Section 3.6.

3.6 OPTIMISATION, NOT DATA, LIMITS THE 4-STEP STUDENT

The 118k runs expose a property of the training recipe rather than of the method: the consistency loss plateaus early and neither more captions nor more updates lower it. Both configurations use a linear warm-up followed by a *constant* learning rate, and the pre-clip gradient norm (130–900) exceeds the clipping threshold of 1 on every update, so every update is a fixed-size normalised step and the weights random-walk on a plateau, which is also why single checkpoints oscillate and why averaging is needed. Under this schedule the loss falls by the same 7% over the first 6,000 updates at 3k and over 57,000 updates at 118k, the per-noise-level losses end at the same values at both scales, and the noisiest supervised state ($k=3$) carries a third of the loss and never improves.

We therefore re-ran both recipes on the 3k pool with a converged schedule – cosine decay to zero, an effective batch of 16 captions (4 GPUs $\times$ 4 accumulated), peak learning rate 10^{-5} and the supervised window narrowed to $\mathcal{K} = \{4, 5, 6, 7\}$ – for 3,000 updates (one seed). The optimisation problem disappears: raw final and averaged checkpoints agree to 0.002, and naive CD alone rises to 0.486 CompBench, above its old averaged score and above every old-schedule 118k student. On this converged baseline our recipe scores 0.488: the edge shrinks from +0.015 to +0.002–0.003, inside single-seed noise, with ours still ahead on shape, numeracy and complex prompts and behind on 2D-spatial ones. The honest reading is that scored selection with the projector reward mostly compensates for weak optimisation; it is not additive with a well-converged schedule at this scale. The same converged schedule at 118k (two passes, 14,244 updates of 16 captions, one seed) gives naive CD 0.4747 and ours 0.4818 on the last-five average (0.4744 and 0.4780 on the raw final checkpoints; GenEval2 21.9 and 22.3): a +0.007 edge for our recipe, between the old-schedule gain and the 3k converged result, and, notably, both 118k students remain below their 3k counterparts trained for sixteen passes – at this compute the distillation is limited by optimisation and by the number of passes over the selected trajectories, not by the number of distinct captions.

3.7 ADDITIONAL ABLATIONS

The following one-factor ablations were run on the 3k pool with the same protocol and are reported in the appendix; only their titles are given here.

- Selection rule: hard selection of the best candidate vs. Boltzmann-weighted vs. sampled vs. uniform candidate weighting.
- Projector refresh schedule: frozen vs. every 100 updates at 4 or 16 steps vs. every 25 updates (Table 4).
- DINOv2 patch pooling: mean vs. max, for selection and reward (Table 4).

Table 8: CompBench by category at 118k, three seeds, last-five-checkpoint averages, one image per prompt.

Category (118k, 3 seeds, last-5 average)	Naive CD	Ours	Ours – naive (p)
Color	0.7755 ± 0.0044	0.7965 ± 0.0029	$+0.0209 \pm 0.0071$ (0.04)
Shape	0.5338 ± 0.0066	0.5515 ± 0.0102	$+0.0177 \pm 0.0168$ (0.21)
Texture	0.7051 ± 0.0063	0.7226 ± 0.0041	$+0.0175 \pm 0.0099$ (0.09)
2D-spatial	0.1971 ± 0.0086	0.2109 ± 0.0113	$+0.0138 \pm 0.0199$ (0.35)
3D-spatial	0.2999 ± 0.0041	0.3190 ± 0.0068	$+0.0190 \pm 0.0109$ (0.09)
Numeracy	0.5483 ± 0.0029	0.5673 ± 0.0061	$+0.0189 \pm 0.0084$ (0.06)
Non-spatial	0.3141 ± 0.0006	0.3126 ± 0.0010	-0.0014 ± 0.0008 (0.10)
Complex	0.3606 ± 0.0030	0.3677 ± 0.0023	$+0.0071 \pm 0.0042$ (0.10)
Mean	0.4668 ± 0.0036	0.4810 ± 0.0050	$+0.0142 \pm 0.0085$ (0.10)

- Consistency-target horizon: 1-step estimate vs. a 2-step chord vs. the trajectory endpoint.
- Training objective: x_0 regression vs. segment-velocity imitation.
- Teacher quality: candidate generation and training with a 16-step and a 28-step teacher.
- Regression onto the reference photograph as a second training target.
- An evolving (exponential-moving-average) teacher in place of the frozen teacher.
- Inference step count: sampling the 4-step student at 2, 8, 16 and 28 steps.

A PROJECTOR PRETRAINING

Algorithm 2 Pretrain the projector P_ϕ (once, before any student training)

Require: 22,000 captions’ cached candidates (z_j , true embedding $e_j = u(\text{Dec}(z_j))$, true score S_j), disjoint from any pool P_ϕ will later score; $\tau=0.04$

- 1: initialise P_ϕ : 5-block conv encoder (widths 64,128,256,512,512) + MLP head (512→1024→768) + ℓ_2 -normalise
- 2: **for** 30 epochs, batches of 32 captions (128 candidates) **do**
- 3: $\hat{e}_j \leftarrow P_\phi(z_j)$; $\hat{S}_j \leftarrow \langle \hat{e}_j, u(x^{\text{ref}}) \rangle$
- 4: $\mathcal{L}_{\text{pre}} \leftarrow \text{mean}_j[1 - \langle \hat{e}_j, e_j \rangle] + \text{KL}(\text{softmax}(S/\tau) \parallel \text{softmax}(\hat{S}/\tau))$ ▷ Equation 5
- 5: AdamW step on ϕ (one-cycle schedule, peak lr 3×10^{-4})
- 6: **end for**
- 7: keep the epoch with the lowest selection regret on 2,000 validation captions; freeze ϕ

The 27,000 captions used here (22,000 training, 2,000 validation, 3,000 test) are disjoint from the 3k training pool’s captions only in their role: the 3,000 test captions *are* the training pool, on which the projector is evaluated but never trained. On them the pretrained projector agrees with the decoded scorer’s choice on 49% of captions and its selection regret, the decoded score of the decoded argmax minus that of the projector’s argmax, is 0.029.

B GENEval2 BREAKDOWN

C ADDITIONAL QUALITATIVE RESULTS

Figures 7 and 8 show an uncurated random draw of prompts (a fixed random seed, no selection on scores). Figures 9 and 10 show the prompts with the largest naive CD-over-ours margin. On the two CompBench prompts the images are nearly identical and the score difference comes from the automatic evaluator (the pig’s snout overlapping the clock face defeats the object detector; the bracelet-and-knife pair is scored by a VQA model that answers differently on near-identical crops); this evaluator sensitivity is the main source of per-prompt noise in both benchmarks. On the two GenEval2 prompts our student makes a genuine counting error, drawing one object too many (eight or nine dogs, five backpacks), which the geometric-mean scoring punishes with a score near zero.

Table 9: GenEval2 by skill and by prompt complexity, 3k pool, seeds 0–2, averaged checkpoints. Overall is the official score (each prompt’s geometric mean over its atoms, averaged over prompts); collapsed is the share of prompts with a geometric mean below 0.05, i.e. at least one atom entirely wrong. Skill rows are the mean score of the atoms tagged with that skill; atom-count rows are the mean score of the atoms of prompts with that many atoms (100 prompts per count).

GenEval2 measure ($\times 100$), 3k, seeds 0–2	Naive CD	Ours	Ours – naive (p)
Overall (official; per-prompt geometric mean)	22.5 ± 0.9	23.5 ± 1.6	$+1.0 \pm 0.9$ (0.20)
Collapsed prompts (%; score < 0.05)	36.6 ± 3.4	33.7 ± 2.1	-3.0 ± 1.4 (0.07)
Skill: object (per atom)	84.2 ± 1.8	84.1 ± 0.8	-0.0 ± 1.2 (0.95)
Skill: count (per atom)	36.2 ± 0.9	36.5 ± 1.1	$+0.3 \pm 1.3$ (0.74)
Skill: attribute (per atom)	70.8 ± 1.1	72.5 ± 1.4	$+1.8 \pm 1.7$ (0.22)
Skill: position (per atom)	47.4 ± 3.7	51.1 ± 2.0	$+3.7 \pm 1.7$ (0.06)
Skill: verb (per atom)	26.1 ± 3.9	24.6 ± 1.0	-1.5 ± 3.4 (0.53)
Atoms = 3 (per atom)	73.7 ± 0.6	73.1 ± 1.1	-0.6 ± 1.5 (0.53)
Atoms = 4 (per atom)	70.2 ± 1.5	72.0 ± 1.2	$+1.8 \pm 0.6$ (0.04)
Atoms = 5 (per atom)	65.5 ± 0.9	66.8 ± 1.8	$+1.3 \pm 1.7$ (0.32)
Atoms = 6 (per atom)	65.2 ± 0.5	65.0 ± 1.9	-0.1 ± 1.5 (0.91)
Atoms = 7 (per atom)	61.6 ± 0.9	60.9 ± 1.0	-0.8 ± 0.2 (0.01)
Atoms = 8 (per atom)	57.4 ± 2.1	58.1 ± 1.6	$+0.7 \pm 2.7$ (0.70)
Atoms = 9 (per atom)	55.4 ± 1.9	58.3 ± 1.2	$+2.9 \pm 1.6$ (0.09)
Atoms = 10 (per atom)	50.6 ± 2.0	51.3 ± 1.6	$+0.7 \pm 3.0$ (0.73)

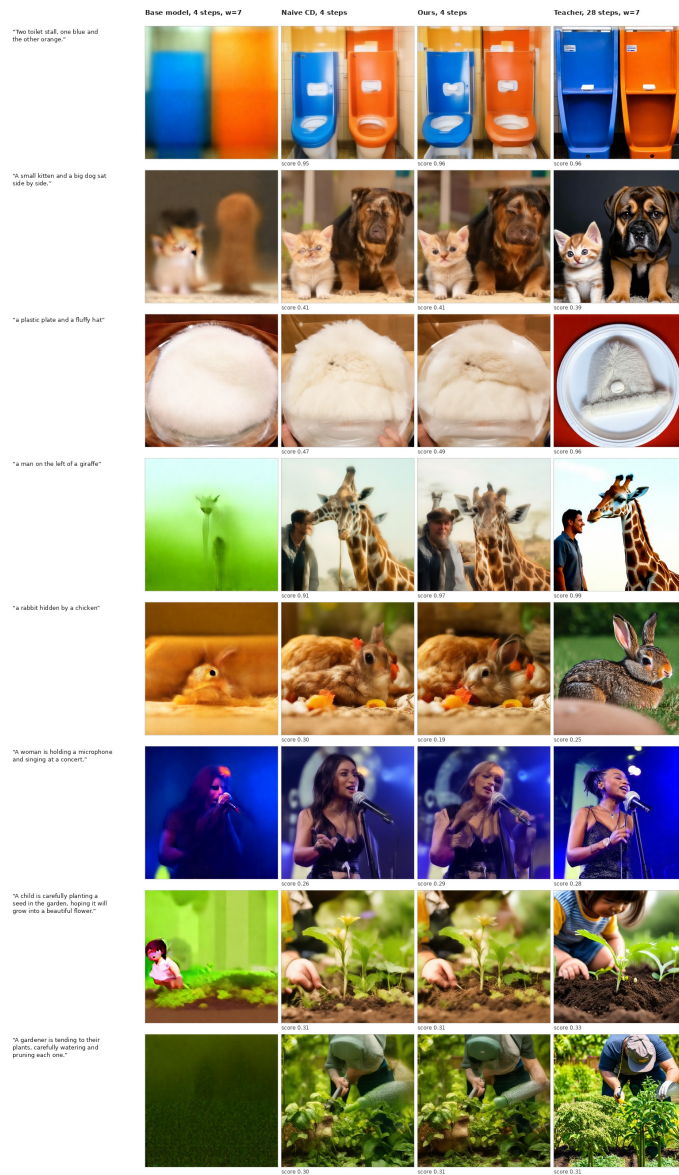


Figure 7: Uncurated T2I-CompBench prompts (eight prompts drawn at random with a fixed seed). Columns share the initial noise; scores are the official per-category evaluator’s.

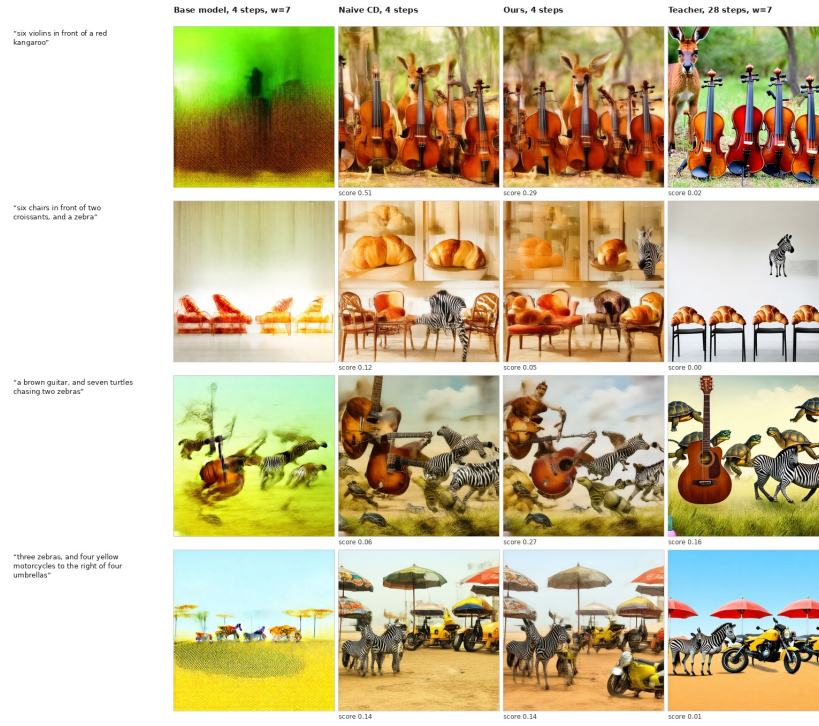


Figure 8: Uncurated GenEval2 prompts (four prompts drawn at random with a fixed seed).

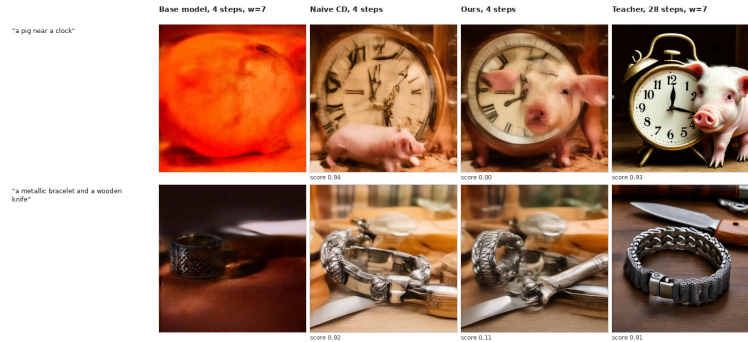


Figure 9: The two T2I-CompBench prompts with the largest naive CD-over-ours score margin.

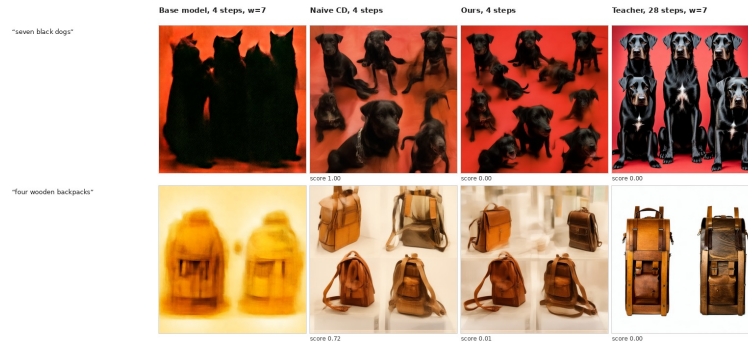


Figure 10: The two GenEval2 prompts with the largest naive CD-over-ours score margin.